VISION Resistivity Dual Frequency Recorded Mode

Schlumberger

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible

FL: Norwegian North Sea
FL1: Section: 26 in.
FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level
Ground Level: 307.00 m

Acquisition Dates:	05-Dec-2019 -- 07-Dec-2019
Log Interval:	341.00(m) -- 932.00(m)
Index Types:	Measured Depth
Index Scales:	1:500
Depth Source:	Driller's Depth
Depth Sensor:	3rd Party Depth
Print Type:	Final
Spud Date:	02-Dec-2019

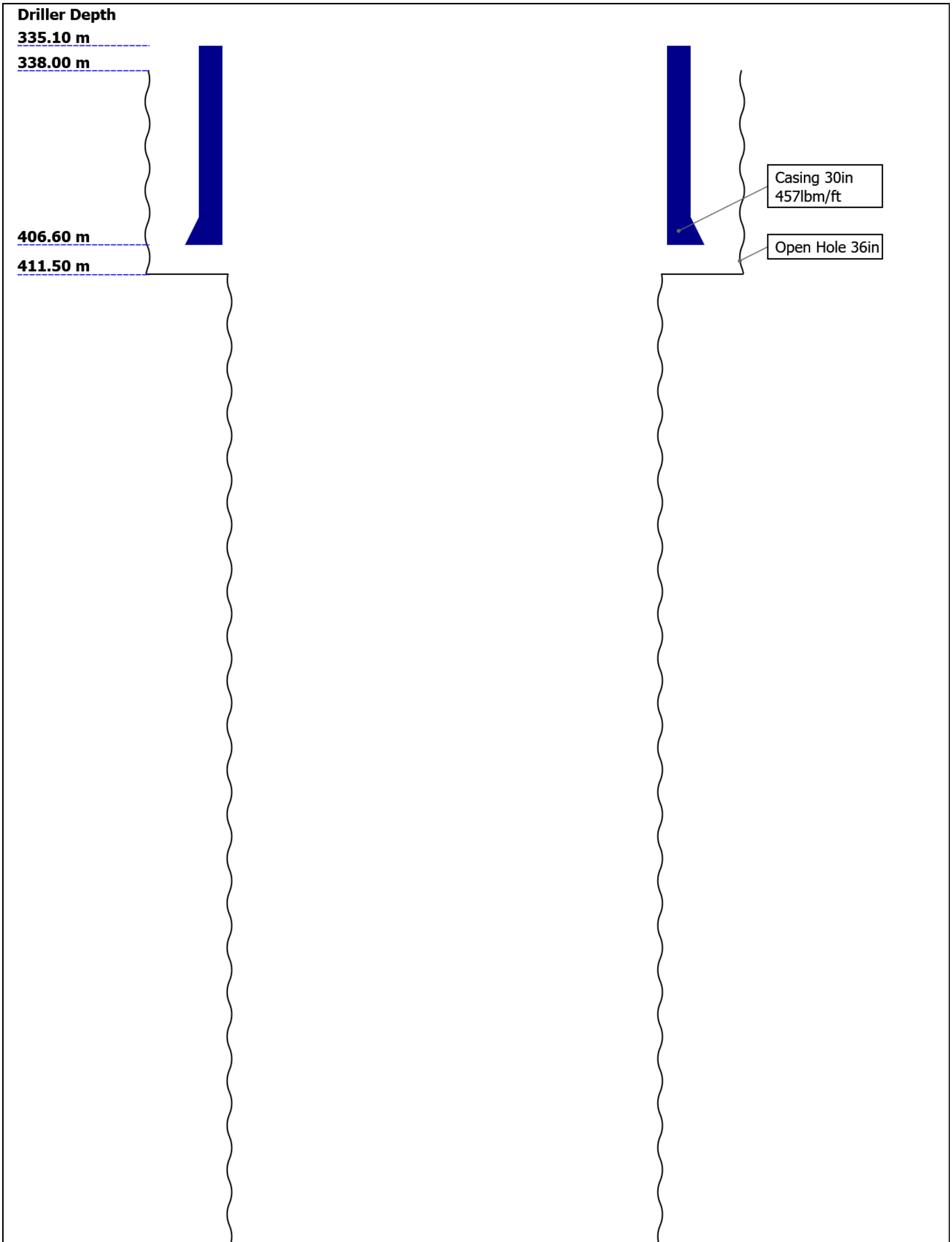
Other Services:
Direction & Inclination (Surveys)
Directional Drilling
Annular Pressure While Drilling

Disclaimer

THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

Contents

Well Sketch



Driller Depth
335.10 m
338.00 m
406.60 m
411.50 m
Casing 30in
457lbm/ft
Open Hole 36in

Borehole Size/Casing Record

Bit							
Bit Size (in)	36	26					
Top Driller (m)	338	411.5					
Bottom Driller (m)	411.5	931					
Casing							
Size (in)	30						
Weight (lbm/ft)	457						
Inner Diameter (in)	27.067						
Grade	X56						
Top Driller (m)	335.1						
Bottom Driller (m)	406.6						

Operational Run Summary

Parameter (unit)	**Run 3**						
Date Log Started	05-Dec-2019						
Time Log Started	01:03:12						
Date Log Finished	07-Dec-2019						
Time Log Finished	02:10:48						
Bit Size (in)	26.000						
Bit Start Depth (m)	411.50						
Bit Stop Depth (m)	931.00						
Top Log Interval (m)	406.60						
Bottom Log Interval (m)	906.80						
Max Hole Deviation (deg)	1.12						
Azimuth of Max Deviation (deg)	120.28						
Logging Unit Number	A615						
Logging Unit Location	Service Office						
Recorded By	J.Johansen						
Witnessed By	S.Vikra						
Service Order Number	19SCA0085						

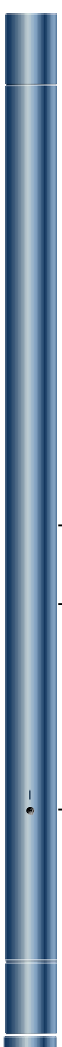
Borehole Fluids

Parameter(unit)	Run 3						
Fluid Type	Water						
Fluid Name	Seawater						
Max Recorded Temperatures (degC)	16						
Source of Sample	Active Tank						
Salinity (ppm)	33579.09						
Density (g/cm3)	1.03						
Funnel Viscosity (s)							
Fluid Loss (cm3)							
PH							
Source RMF							
RMC	Pressed						
RM @ Meas Temp (ohm.m@degC)	0.19 @ 23.9						
RMF @ Meas Temp (ohm.m@degC)	0.19 @ 23.9						
RMC @ Meas Temp (ohm.m@degC)							
RM @ BHT (ohm.m@degC)	0.23 @ 16						
RMF @ BHT (ohm.m@degC)	0.23 @ 16						
RMC @ BHT (ohm.m@degC)	NaN @ 16						
Total Solid (%)							
High Gravity Solids (%)							

Remarks and Equipment Summary

Run 3: Toolstring

Equip name	Length	MP name	Offset
TELE900:G5093	**35.13**	TeleScope	
MSSU900:71539			
Upper Extender:26-02-16			
MDC900:G5093			
MMA:2068			
MDI:3956			
PMGR:455			
PMEA:3708			
MTA:2526			
MTK900:008/298895-2142			
MSSD900:973247-1			
Lower Extender:12/142			
		D&I	**30.56**
		GR	**29.91**
		ROP	**28.21**
ARC9:5969	**26.45**	arcVISION	



Run 3: Remarks

All depths referenced to driller's depth. See EOWR for depth tracking details.

All data from tool memory. All data acquired while drilling.

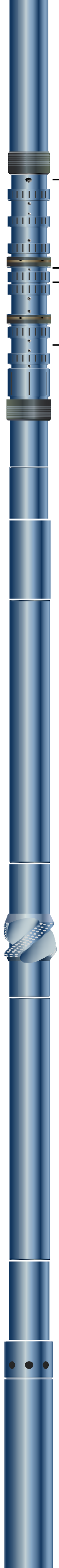
Gamma Ray measurement is corrected for collar thickness only.

Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.

Phase Resistivity 16 and 22 in. spacing adversely affected by large borehole corrections.

Run 3 and 26 in. section TD at 931 m.

ARC9:5969 20.43
ARDC:5969
Upper Extender:0
7-184
CDJA
AREA:8374
APWD:1820
ARSS:105209
Lower Extender:
13/013



X/O: 9 1/2":10 5983 **20.68** X-Over

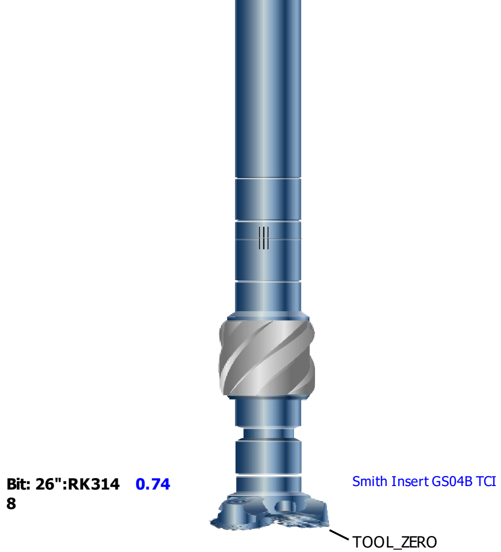
NMDC: 9 1/2"[2]:78421-2 **19.9** Non-Mag Pony Collar

Stab: 9 1/2":M WS 4018 **16.93** 25 3/4 in. Non-Mag Stab

NMDC: 9 1/2"[1]:26105 **13.99** Non-Mag Pony Collar

Flt Sub: 9 1/2" :SDT-9316 **11.52** Float Sub

Motor: 11 1/4" :3203 **9.19** PowerPak



Lengths are in m
Maximum Outer Diameter = 26.000 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 3

VISION Resistivity - Dual Frequency

Software Version

Acquisition System	Version
Maxwell 2019.2	9.2.113335.3100
Application Patch	DnM_NPD-KAI-2019.2_9.2.114235

Computation	Description	Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.2.113335.3100
ARC9GammaRayComputation	ARC9 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.2.113335.3100

Tool Interface	System Version	Loaded Version
HSPM	2019.0c.001	9.2.113335.3100

Tool Elements	Description	Software Version	Firmware Version
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 9.00 Inch Tool Drilling Collar	9.2.113335.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 3	Drilling	Down	33.15 m	930.00 m	05-Dec-2019 1:03:12 AM	07-Dec-2019 2:10:48 AM	Yes

All depths are referenced to toolstring zero

Log

Company:Equinor Well:31/5-7
Run 3: Drilling:S010

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 12-Dec-2019 15:13:58

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM
0.2 ohm.m 200

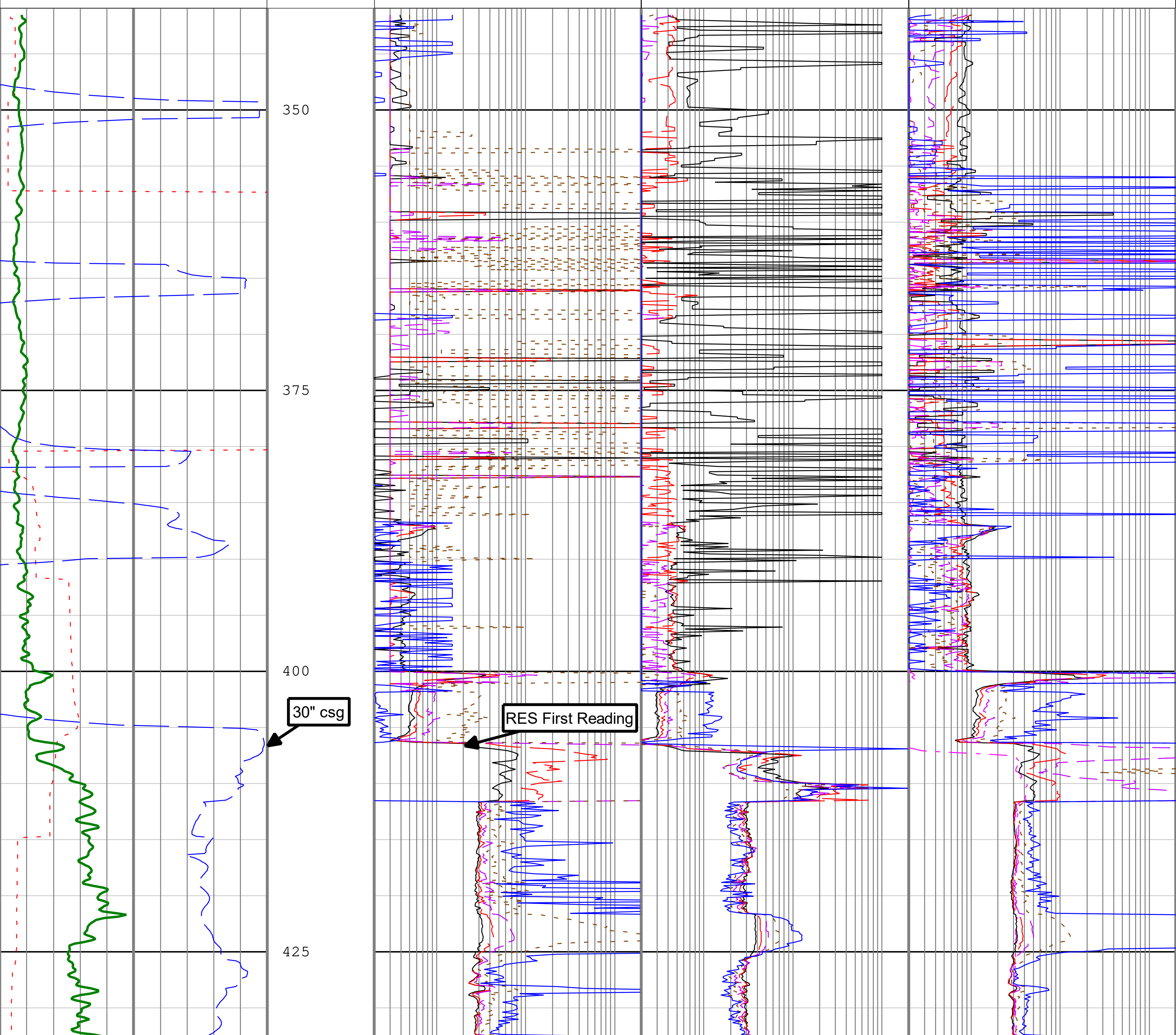
Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM
0.2 ohm.m 200

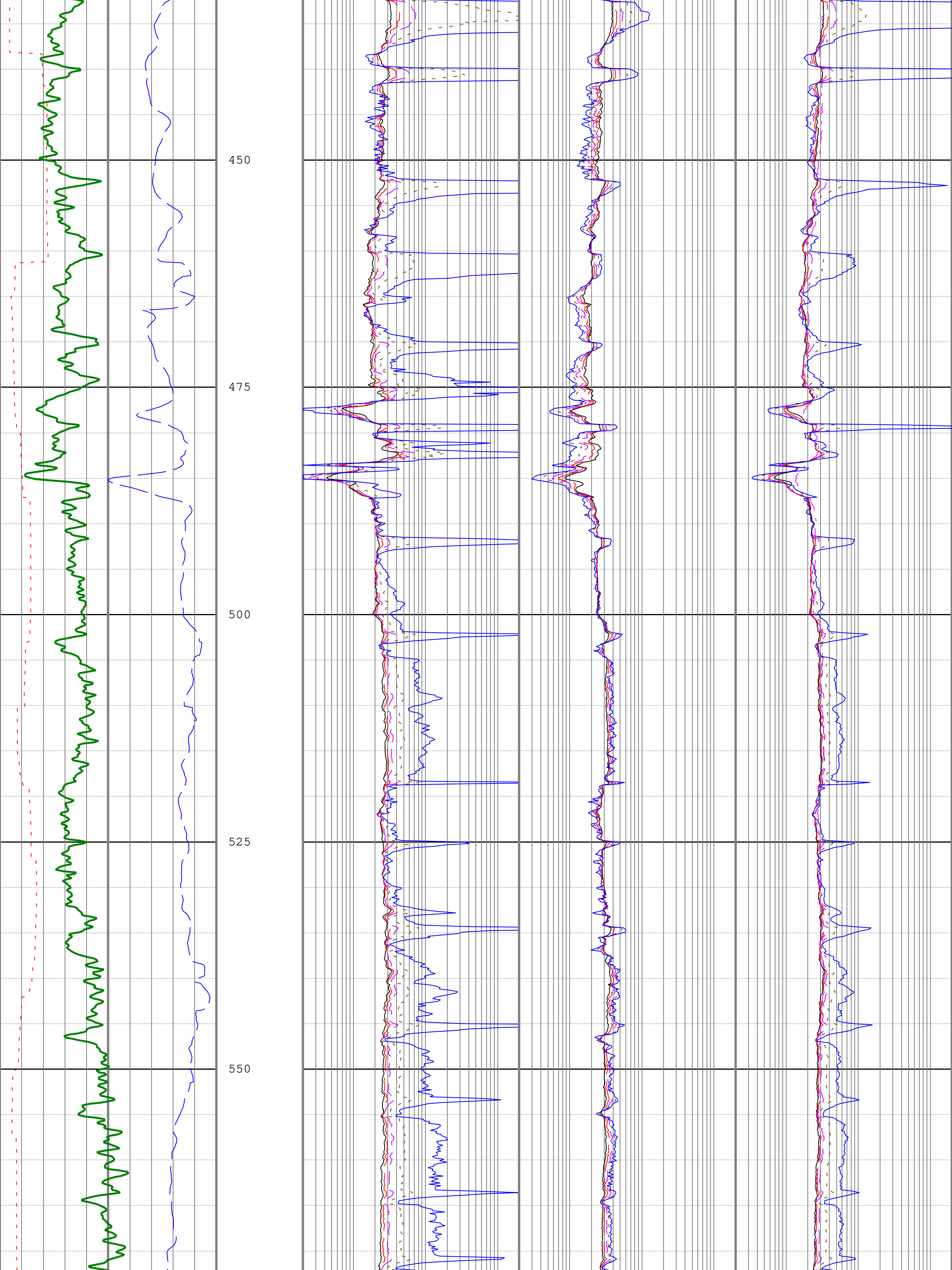
Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM

Phase Shift Resistivity 34 inch

Track 1		Track 2	Track 3	Track 4
		0.2 ohm.m 200	0.2 ohm.m 200	Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM 0.2 ohm.m 200
		Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM 0.2 ohm.m 200
Resistivity Time After Bit (ARTM) ARC9 0 h 10		Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2 ohm.m 200
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT 200 m/h 0		Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM 0.2 ohm.m 200
Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM 0 gAPI 150		Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM 0.2 ohm.m 200	

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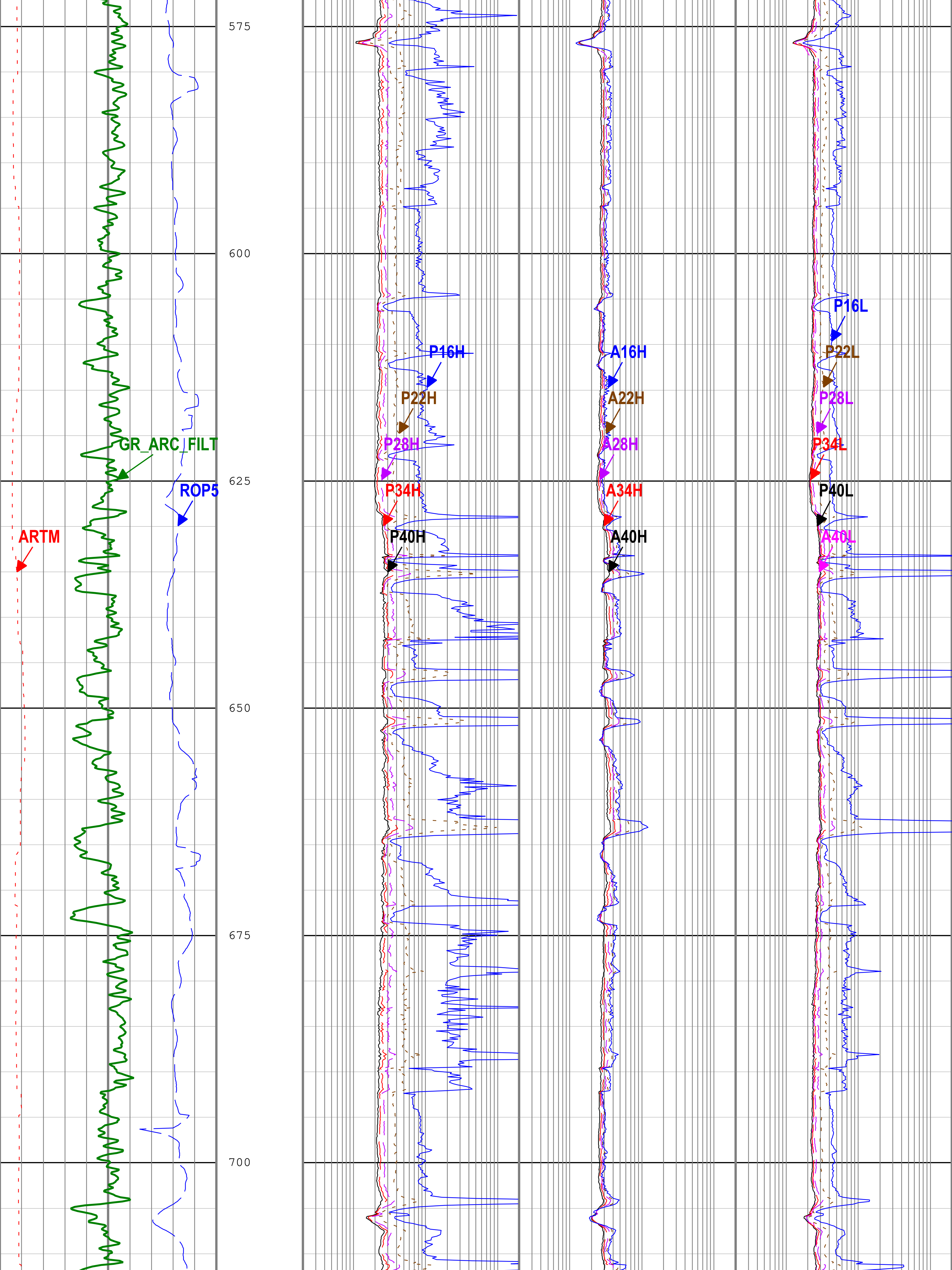
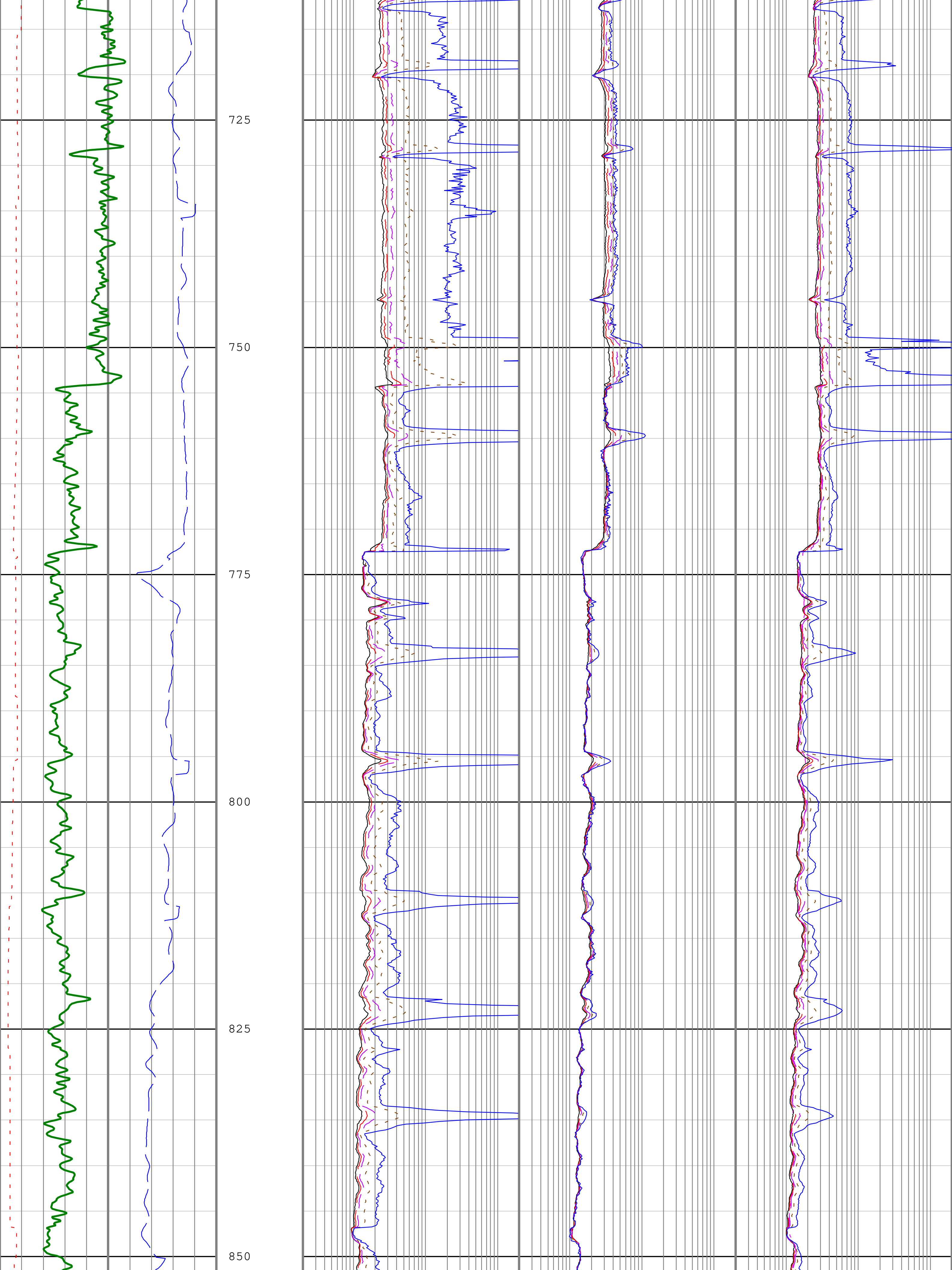
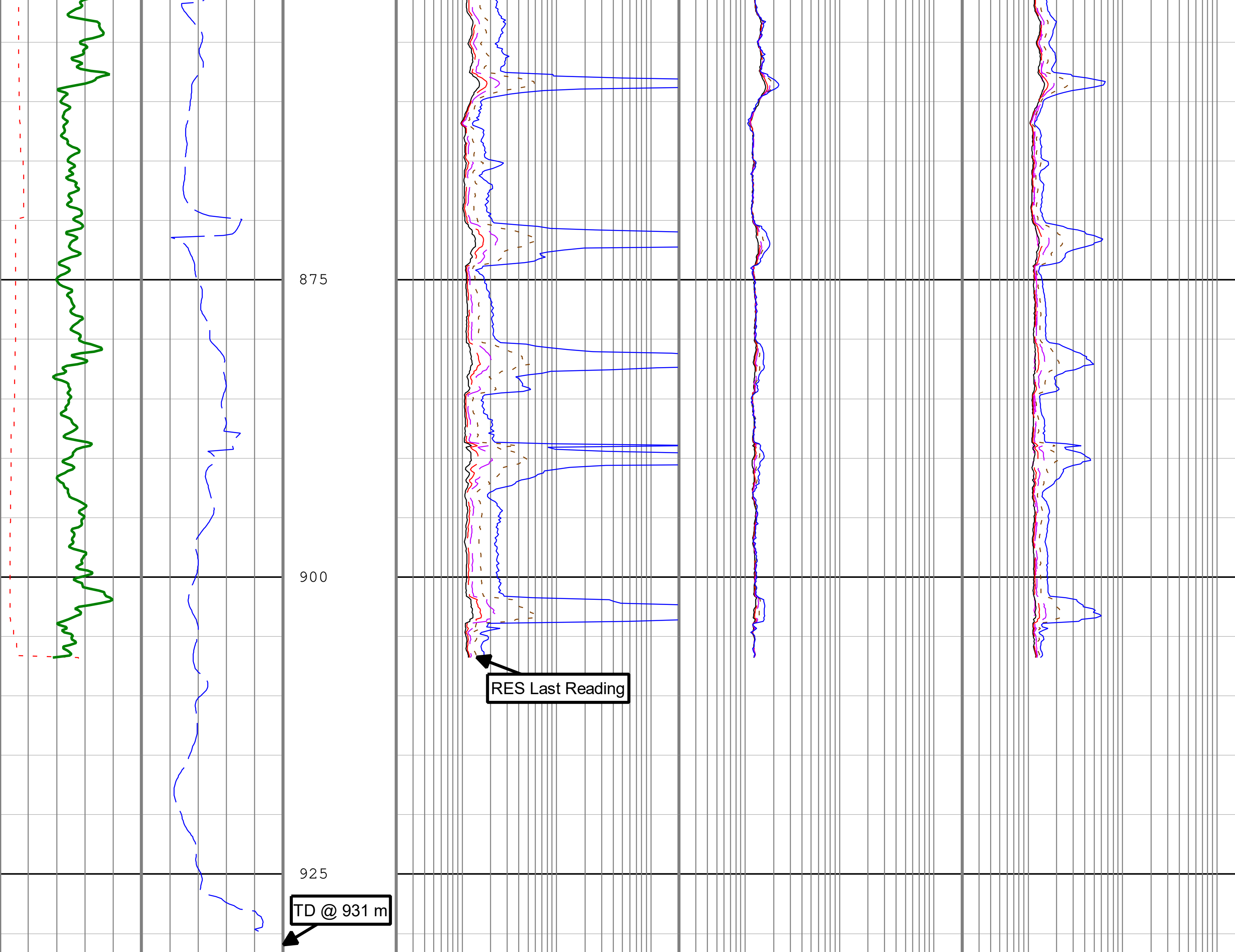
475
500
525
550


575
600
625
650
675
700
GR_ARC_FILT
ROP5
ARTM
P16H
P22H
P28H
P34H
P40H
A16H
A22H
A28H
A34H
A40H
P16L
P22L
P28L
P34L
P40L
A40L


725
750
775
800
825
850



Resistivity Time After Bit (ARTM) ARC9
0 h 10

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT
200 m/h 0

Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM
0 gAPI 150

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM

0.2 ohm.m 200

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 12-Dec-2019 15:13:58

Channel Processing Parameters

Run 3: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	0	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	1.03	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.19	ohm.m

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
BS	36	341	411.5
BS	26	411.5	931

All depth are actual.

Tool Control Parameters

Calibration Report

ARC9 (Array Resistivity Compensated 900) Calibration - Run 3

Primary Equipment :

High Pressure without AIM Receiver AREA 8374

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 09:51:54 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.353	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.149	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.129	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.142	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.746	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	0.713	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-0.663	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	0.666	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-0.695	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	0.651	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.523	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	5.988	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.294	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	3.975	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.915	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.360	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.415	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.390	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.422	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.384	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:35:50 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Schlumberger

VISION Resistivity

Dual Frequency

Recorded Mode